

**Priming with Instagram: Understanding the Relationship Between Self-Perception and
Social Media**

Everett tai

Psyc 11, Dartmouth College

Dr. Clark, Department of Psychology

27 August 2024

Introduction:

Social media is an exponentially growing platform that has made a significant impact on the modern world. Platforms such as Instagram have transformed the way individuals engage with each other, offering unprecedented opportunities for social interaction and self-presentation. By sharing snippets of content from their personal lives through images, videos, and much other information, users become immersed in environments where they are constantly exposed to social comparisons, which can have profound effects on self-esteem and feelings of social support. As the number of users continues to grow, self-perceptions in response to social media interaction and activity have become an area of concern across many research studies.

Leon Festinger's (1957) social comparison theory suggests that individuals have an inherent tendency to evaluate themselves by comparing their abilities, achievements, and attributes with those of others. Further research has shown that upward social comparisons – where individuals compare themselves to others who appear better off – can lead to lower self-esteem and increased feelings of inadequacy (Vogel et al., 2014). In a virtual environment where people can fabricate the outlook on their lives, these comparisons have vast implications on the current state of self-perceptions in social media users today. In a study by Abulla Alblooshi, researchers found that people with higher self-esteem tend to take more selfies than those with lower self-esteem (Alblooshi, 2015). Additional research conducted by Kristina Miljeteig and Tilmann von Soest explores the longitudinal relationship between social media and self-esteem, finding that lower stability in self-esteem, characterized by greater fluctuations in self-esteem, predicted more frequent social media use (Miljeteig et al., 2022).

While existing research highlights the role of social comparisons in self-esteem, less is known about how specific priming of social versus nonsocial contexts influences these effects in

real-time interactions on social media platforms like Instagram. In this study, we aim to build on existing research by examining the impact of social media based social comparisons on self-esteem, utilizing a combination of correlational and experimental approaches. Particularly, we are interested in investigating the impact of social media by examining how priming individuals to think about social versus nonsocial activities and the order of exposure to social versus nonsocial context influences their self-esteem, social engagement, and perceptions of loneliness when interacting with Instagram content. By exploring the mechanisms through which social media influences self-perception, this research seeks to provide a deeper understanding of the psychological effects of social media use, with implications for both individual well-being and broader social dynamics. In our experiment, participants were divided into a nonsocial group, primed to think about solitary activities, and a social group, primed to think about social interactions with friends. Following the priming, participants were assessed on their self esteem three times – once before viewing social media images, once after viewing the first post, and again after the second post. Participants were additionally assessed on their perceptions of social support and their likelihood of engagement with the posts. At the end of the study, participants were asked to provide screenshots of their Instagram pages in order for the researchers to collect data on number of followers, following, and types of posts people were posting to identify any correlations with participants' responses. Based on previous findings, we hypothesized that people in the social group would feel more socially supported and have greater self-esteem, with the number of followers and following as well as the types of posts people display also being correlated to self-esteem.

Method

Participants

Twenty-eight Dartmouth Psychology 11 students and faculty participated in this study (n=28). Using an online random generator, participants were randomly assigned into one of two priming groups: a social priming group (Group 1) or a non-social priming group (Group 2). The sample consisted of participants who attended class and consented to participate in the study, with uneven group sizes (13 and 15) due to absences during the data collection period.

Materials and Measures

The Rosenberg Self-Esteem Scale (RSES), a 10-item measure with responses on a 4-point Likert scale ranging from ‘Strongly Disagree’ to ‘Strongly Agree,’ was used to assess participants’ self-esteem on three separate occasions (Rosenberg, 1965). Additionally, loneliness was measured using a scale we developed for this study, with items tailored to assess feelings of loneliness and social support that participants rated their feelings for on a semantic differential scale from 0-100. This scale was designed to capture participants' perceived social support and feelings of loneliness following the priming task.

Design

This study employed a mixed factorial design with one between-groups factor (priming condition: social versus non-social) and one within-groups factor. The between-groups factor was manipulated by random assignment of participants to either Group 1 or Group 2. The within-groups factor involved measuring self-esteem and loneliness after the presentation of social media posts. Additionally, we collected screenshots of their Instagram profiles to gather metrics such as followers, following, # of posts, and type of posts (selfies or group photos) to analyze with participants self-esteem and loneliness assessments.

Procedure

Participants were first provided with a consent form and were sent a specific link via email corresponding to their group assignment. Participants were informed that the study aimed to explore how different types of social activities might influence their perceptions of social media content. They were instructed to complete the priming task and subsequent assessments as honestly as possible. Participants in Group 1 were asked to list 5 activities they did with friends in the past two weeks. Participants in Group 2 were asked to list 5 activities they did alone in the past two weeks. Upon completion, participants completed the full RSES.

After the priming task and first self-esteem assessment, participants were shown a post from Instagram. The order of post presentation was counterbalanced: participants in Group 1 (social priming) first viewed the social post (a selfie of a group of friends) followed by the non-social post (a book in the grass), while participants in Group 2 (non-social priming) first viewed the non-social post followed by the social post. This counterbalance was done in an effort to maintain the possible effects of priming. Participants completed a derivative of the RSES along with a loneliness assessment after viewing the first post, and again after viewing the second post. After completing these tasks, participants uploaded their social media profiles and were subsequently debriefed on the study's purpose and manipulations.

For the Instagram analysis, we evaluated the first six thumbnails of the posts participants provided in their screenshots. If a participant had more than six posts, we only analyzed the first six; if they had fewer than six, we analyzed whatever was available. Specifically, we looked at the number of group photos and the number of selfies on their page. Additionally, we collected data on each participant's number of followers and number of people they follow.

Statistical Analysis

Independent samples t-tests were conducted to determine the effect of priming group (social versus non-social) on total self-esteem scores. A repeated measures ANOVA was performed to examine the effect of the type of post presented on loneliness ratings within participants. A correlation matrix was constructed, and Pearson correlation coefficients were calculated to explore the relationships between participants' Instagram metrics and self-esteem.

Results

Hypothesis 1 - Priming on Self-Esteem

To examine the first hypothesis that participants in the social priming group will have a higher self-esteem (RTotal) than those in the non-social priming group, we used an Independent Samples T-test to compare RTotal scores across the two priming groups. The independent variable was the priming condition (social vs. nonsocial), and the dependent variable included self-esteem scores.

Independent Samples T-Test

Independent Samples T-Test

		Statistic	df	p
RTotal	Student's t	1.26	26.0	0.220

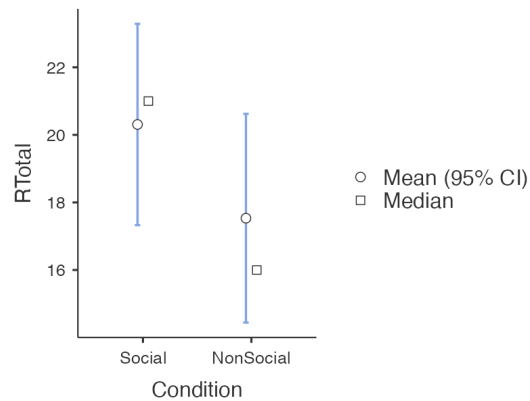
Note. $H_a: \mu_{\text{Social}} \neq \mu_{\text{NonSocial}}$

Group Descriptives

	Group	N	Mean	Median	SD	SE
RTotal	Social	13	20.3	21.0	5.48	1.52
	NonSocial	15	17.5	16.0	6.10	1.58

Plots

RTotal



The RTotal was not significantly different between the 13 participants in the social priming group ($M = 20.3$, $SD = 5.48$) and the 15 participants in the non-social priming group ($M = 17.5$, $SD = 6.10$) as revealed by an Independent Samples T-Test ($t(26) = 1.26$, $p = .220$).

The current results are not significant and thus we fail to reject the null hypothesis; we do not have sufficient evidence to support the hypothesis that RTotal would be different between the social and non-social priming groups.

Hypothesis 2 - Type of Post and Loneliness

To test the second hypothesis that the type of post participants viewed would have a main effect on loneliness scores, we used a One-Way Repeated Measures ANOVA comparing the loneliness scores between all 28 participants.

Repeated Measures ANOVA

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Post Manipulation	407.70	1	407.70	1.5056	0.231	0.055
Post Manipulation * Condition	7.70	1	7.70	0.0284	0.867	0.001
Residual	7040.51	26	270.79			

Note. Type 3 Sums of Squares

[3]

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Condition	69.3	1	69.3	0.0678	0.797	0.003
Residual	26564.6	26	1021.7			

Note. Type 3 Sums of Squares

Post Hoc Tests

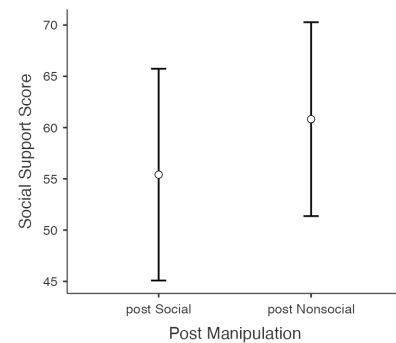
Post Hoc Comparisons - Post Manipulation

Comparison		Mean Difference	SE	df	t	Ptukey
Post Manipulation	Post Manipulation					
post Social	- post Nonsocial	-5.41	4.41	26.0	-1.23	0.231

[4]

Estimated Marginal Means

Post Manipulation



Estimated Marginal Means - Post Manipulation

Post Manipulation	Mean	SE	95% Confidence Interval	
			Lower	Upper
post Social	55.4	5.02	45.1	65.7
post Nonsocial	60.8	4.60	51.4	70.3

The type of post the participants viewed had no significant effect on loneliness scores as revealed by a Repeated Measures ANOVA, $F(1, 26) = 1.5056$, $p = 0.231$, $\eta^2_p = 0.055$ with 28 participants. The estimated marginal means table demonstrates that social support scores increased with viewing nonsocial posts. The specific scores are in the above table and are as follows:

- Post Social: $M = 55.4$, $SE = 5.02$
- Post Nonsocial: $M = 60.8$, $SE = 4.80$

Therefore, these results do not support the hypothesis that loneliness scores change with the type of social media post viewed.

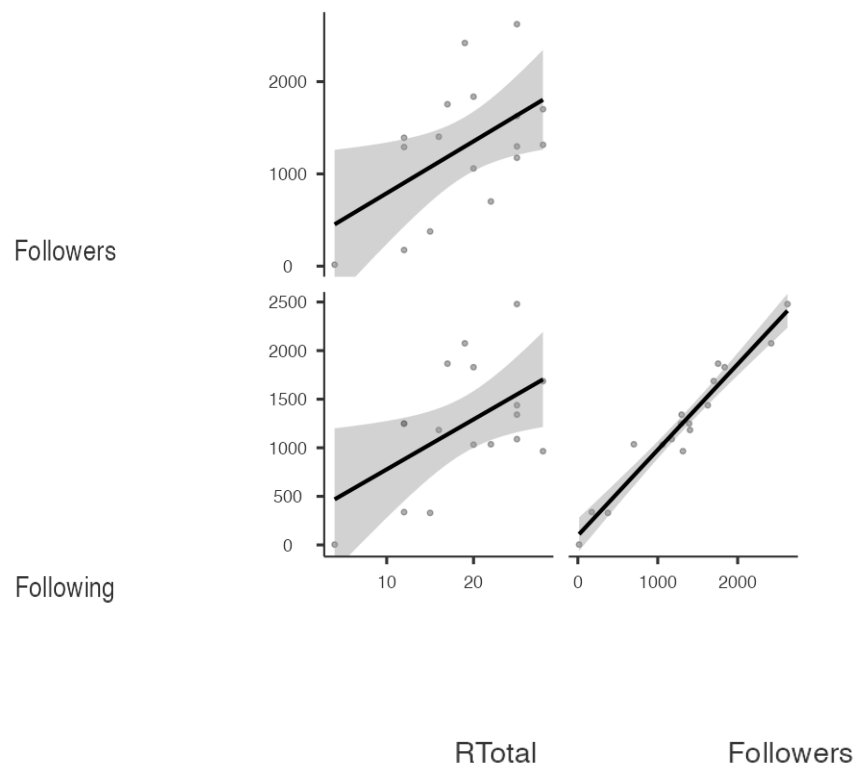
Hypothesis 3 - Self Esteem and # of Followers/Following

To examine the third hypothesis that RTotal positively correlates with the number of followers and following, we used a correlational matrix to see if there is any linear relationship between RTotal and the number of followers/following on Instagram.

Correlation Matrix

Correlation Matrix

		RTotal	Followers	Following
RTotal	Pearson's r	—		
	df	—		
	p-value	—		
Followers	Pearson's r	0.534	—	
	df	15	—	
	p-value	0.027	—	
Following	Pearson's r	0.537	0.972	—
	df	15	15	—
	p-value	0.026	<.001	—



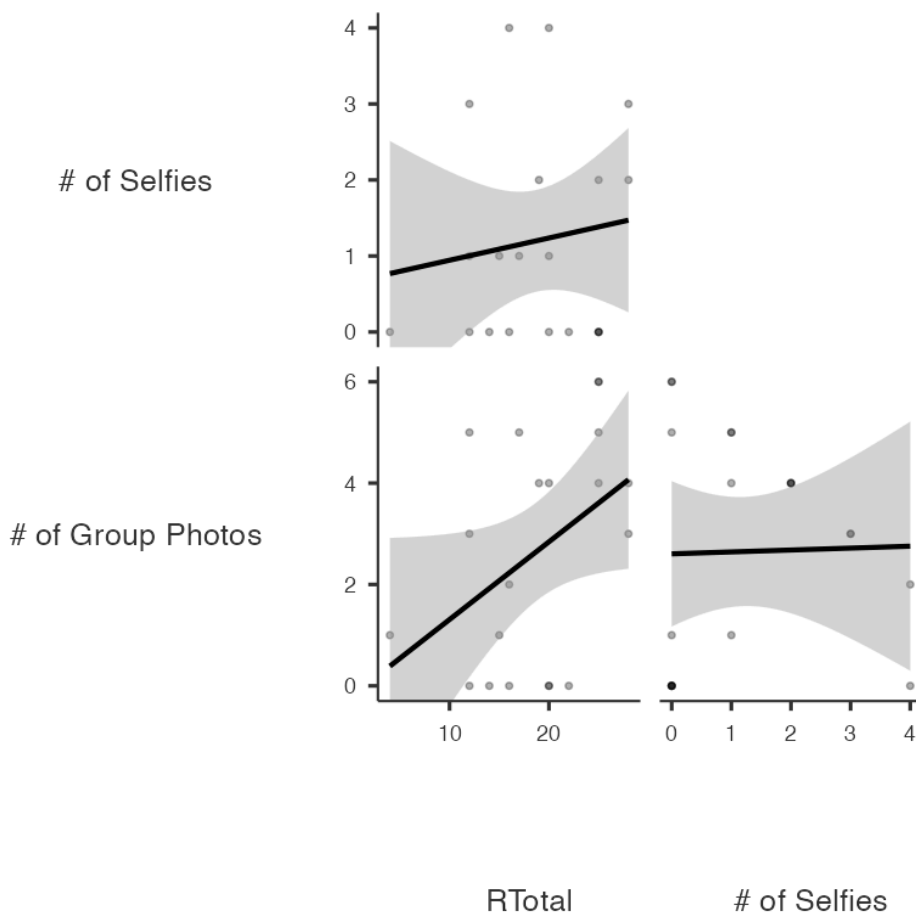
There was a significant positive correlation between the number of followers/following and the self-esteem scores $r(15) = 0.534$, $p < 0.05$ and $r(15) = 0.537$, $p < 0.05$, meaning that a higher RTotal is associated with higher number of followers and following.

Hypothesis 4 - Self Esteem and # Group of Photos/Selfies

To test the fourth hypothesis that RTotal has a positive correlation with number of selfies posted and negative correlation with group photos, we used a correlational matrix that generates a scatter plot.

Correlation Matrix

Correlation Matrix		RTotal	# of Group Photos	# of Selfies
RTotal	Pearson's r	—		
	df	—		
	p-value	—		
# of Group Photos	Pearson's r	0.432	—	
	df	18	—	
	p-value	0.057	—	
# of Selfies	Pearson's r	0.132	0.024	—
	df	18	18	—
	p-value	0.580	0.921	—



RTotal was not significantly positively correlated to # of Selfies ($r(18) = 0.132$, $p = 0.580$), and RTotal was not significantly negatively correlated to # of Group Photos ($r(18) = 0.432$, $p = 0.057$) as revealed by the scatter plot and correlation matrix. Interestingly, as demonstrated in the plot, there is a positive correlation between RTotal and # group photos that are nearly significant, potentially signaling the experiment lacks the power to detect a statistically significant effect.

The current results are not significant and thus we fail to reject the null hypothesis; we do not have sufficient evidence to support the hypothesis that there is a positive correlation between RTotal and # of selfies.

Discussion:

In this study, we sought to investigate the impact of social versus nonsocial priming on self-esteem, social engagement, and perceptions of loneliness when interacting with Instagram content. In doing so, our goal was to shed light on the psychological mechanisms underlying social media use and its effects on well-being. We predicted that several main factors of social media would affect self-esteem and loneliness ratings. First, we predicted that self-esteem scores would be lower in the social priming group than the nonsocial priming group. We also predicted that the type of Instagram post shown to the viewers would have an impact on their loneliness scores. Finally, we anticipated that the number of followers and following on an individual's Instagram pages would correlate with the stability of self-esteem scores.

Though not statistically significant, this study resulted in findings that moved in the correct direction that support our hypotheses, as the social priming group displayed higher self-esteem scores, social support scores were higher with the social instagram post, and

Strengths:

This study was designed with strong construct validity. The measurement of self-esteem was well represented using a well-established scale designed by Morris Rosenberg called the Rosenberg Self-Esteem Scale to assess self-esteem. By ensuring that the RSES was appropriately adapted for our experimental design, we effectively measured the construct of self-esteem in a way that aligns with the theoretical framework of our study. This study also demonstrated strong internal validity, as the experimental design used random assignment to the two different priming conditions, which helped control for confounding variables, ensuring that any observed differences in outcomes can be attributed to the experimental manipulation rather than other

factors. Additionally, counterbalancing was used in the study to minimize order effects of viewing the social or the nonsocial post first. Participants in the social group viewed the social post before the nonsocial post, and participants in the nonsocial group viewed the nonsocial post before the social post.

Limitations and Future Research:

Though the findings of this study were mostly statistically insignificant, there were multiple factors that should be taken into account when assessing the results of this study. One limitation of the present study was that the sample size was relatively small. With only 28 participants in total and several absences, groups only contained 18 (social) and 15 (nonsocial) participants. It might be beneficial to implement the study with more participants, where it would be feasible to create four separate groups, splitting both the priming conditions and the type of post viewed by the participants. Furthermore, it might be more accurate to display more posts to participants, mimicking a more realistic scroll through an Instagram feed. In the data above, the correlation results were moving in the right direction, so with more participants there may or may not be a significant result.

Another limitation of this study stems from an outlier from the dataset. One significant result we obtained was the correlation between the ratio of follower and following to self-esteem scores. One participant in particular scored a R_{Total} of 4. This is far from the usual range of 15-25, indicating extremely low self-esteem. This outlier skews the results of the study, pulling everything further down. When running the statistical test without this outlier, the result was insignificant. However, it is unknown whether this trend would continue or if the outlier would have a big enough impact if there were more participants.

Lastly, the results of this study could be skewed due to the fact that it was drawn from a convenience sample. Each participant in the study was affiliated with the same psychology class at Dartmouth College, meaning that the majority of the sample fell between the age range of 19-22 years old. Although it is common that younger audiences are more involved with social media, it would be more accurate to study a random representative sample of social media users. With a larger, more representative sample size, the results of the study would be more valid and generalizable, allowing us to find more conclusive findings about the role of social priming and social media engagement with shaping self-esteem, social support perceptions, and social behavior. By demonstrating how priming individuals to think about social versus nonsocial activities influences their responses to social media content, we provide valuable insights into the psychological processes underlying social media use. These findings underscore the importance of understanding the contextual factors that influence social media interactions and suggest avenues for promoting healthier and more mindful social media use.

References

- Alblooshi, A. (2015). Self-esteem levels & selfies: The relationship between self-esteem levels and the number of selfies people take and post, and the uses and gratifications of taking and posting selfies (Master's thesis, Middle Tennessee State University).
- Festinger, L. (1957). Social comparison theory. *Selective Exposure Theory*, 16(401), 3.
- Miljeteig, K., & von Soest, T. (2022). An experience sampling study on the association between social media use and self-esteem. *Journal of Media Psychology*.
- Serra, L., & Campaniço, M. (2024). The use of social media: the mediating effect of the number of followers on the relationship between life satisfaction and users' self-esteem.
- Tinakon, W., & Nahathai, W. (2012). A comparison of reliability and construct validity between the original and revised versions of the Rosenberg Self-Esteem Scale. *Psychiatry investigation*, 9(1), 54.
- Vogel, E. A., Rose, J. P., Roberts, L. R., & Eckles, K. (2014). Social comparison, social media, and self-esteem. *Psychology of popular media culture*, 3(4), 206.